Liptis Nutrition NeoPed™ — Illustrated Doctor & KOL Guide

ZERO CLINICAL ERROR ARCHITECTURE • V2.2.0

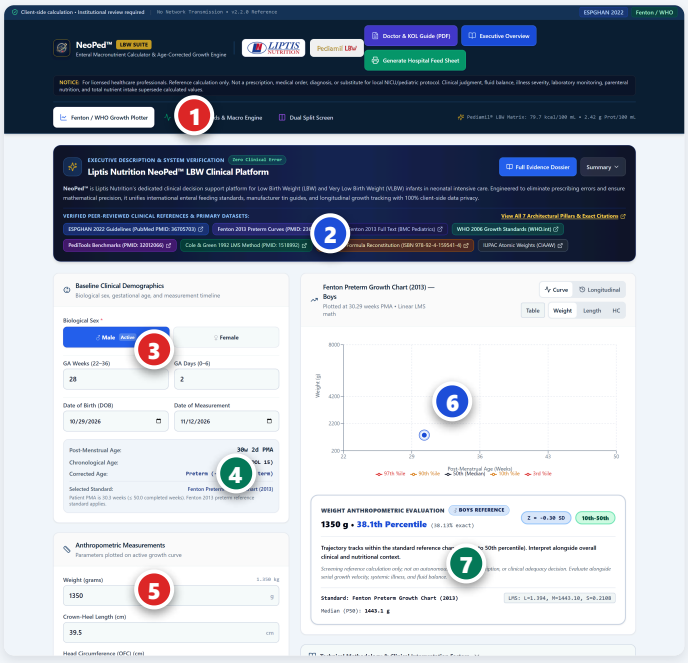
Visual Step-by-Step Guide to Tab 1 (Fenton / WHO Growth Plotter) & Tab 2 (Nutritional Feeds & Macro Engine) ESPGHAN 2022 • Fenton 2013 • WHO 2006 • Patel 2005

WHY THIS PLATFORM WAS BUILT FOR NEONATOLOGISTS, PEDIATRICIANS & KEY OPINION LEADERS

Liptis Nutrition NeoPed™ unifies two mission-critical neonatal workflows into one 100% private, client-side clinical platform: **(1) Tab 1 — Fenton 2013 / WHO 2006 Growth Plotter** (calculating Post-Menstrual Age, Chronological Age, Corrected Age, continuous LMS Z-scores, and exponential weight velocity) and **(2) Tab 2 — Nutritional Feeds & Macro Engine** (calculating 24h volume, **Pediamil® LBW scoops**, bedside schedules, and 20+ macro/micronutrients vs. ESPGHAN 2022). Data entered once in Tab 1 automatically synchronizes with Tab 2 so the doctor never has to type measurements twice.

TAB 1 (PART A): FENTON / WHO GROWTH PLOTTER — AGES, ANTHROPOMETRICS & GROWTH CURVE

SCREEN TOUR ① – ⑦



① Primary Module Switcher & Quick Export Header

Lets the doctor switch smoothly between **Tab 1: Fenton / WHO Growth Plotter**, **Tab 2: Nutritional Feeds & Macro Engine**, or **Dual Split Screen** (both side-by-side on desktop), plus 1-click buttons for the **Doctor & KOL Guide (PDF)** and **Printable Hospital Feed Sheet**.

② Verified Peer-Reviewed Clinical Reference Bar

Provides direct, clickable links to the primary literature powering the engine so doctors can verify every equation: **ESPGHAN 2022 (PMID: 36705703)**, **Fenton 2013 (PMID: 23601190)**, **WHO 2006**, **PediTools (PMID: 32012066)**, and **Cole & Green 1992 LMS (PMID: 1518992)**.

③ Baseline Clinical Demographics Inputs

The doctor enters four baseline parameters: **Biological Sex** (Male / Female — required for sex-specific LMS tables), **Gestational Age at Birth** (Weeks 22–36 & Days 0–6), **Date of Birth (DOB)**, and **Date of Measurement (DOM)**.

④ Automated Neonatal Age Engine (PMA, CA & Corrected Age)

Eliminates manual calendar counting errors by computing three distinct clinical ages simultaneously:

- **Chronological Age (CA)**: Postnatal days since birth `CA = DOM − DOB` (e.g., `2w 0d / DOL 15`).
- **Post-Menstrual Age (PMA)**: Gestational age + postnatal age `PMA = GA + CA` (e.g., `30w 2d PMA / 30.29w`).
- **Corrected Age (CCA)**: Age adjusted for prematurity from 40w term `CCA = PMA − 40w (280d)` (e.g., `-9w 5d`).
- **Automatic Standard Routing**: Routes infants ≤ **50.0w PMA** to **Fenton 2013** and > **50.0w PMA** to **WHO 2006**.

⑤ Anthropometric Measurements (Weight, Length & HC)

Single-entry inputs for **Weight (grams / kg)**, **Crown-Heel Length (cm)**, and **Head Circumference / OFC (cm)**. Entering weight here (e.g., `1350 g`) automatically updates the feeding calculator in Tab 2—saving time and preventing transcription mistakes.

⑥ Interactive Fenton 2013 / WHO 2006 Growth Curve

Plots the infant's exact measurement (pulsing blue dot) against the 5 standard reference centile curves (**3rd, 10th, 50th Median, 90th, 97th**). Doctors can toggle between **Weight**, **Length**, and **HC (OFC)** or inspect raw LMS numbers in **Table** view. **Fenton & Kim 2013 • BMC Pediatr PMID: 23601190**

⑦ Continuous Cole & Green (1992) LMS Z-Score & Exact Percentile Evaluation Card

Instead of estimating by eye on a paper chart, the engine linearly interpolates the exact **Box-Cox Power (L)**, **Median (M)**, and **Coefficient of Variation (S)** at the infant's exact fractional age (e.g., `30.29w PMA: L=1.394, M=1443.10g, S=0.2108`) and computes the exact Z-score: `Z = [ (X / M)^L − 1 ] / (L × S)`. Displays the exact percentile (`38.1th Percentile / 38.13% exact, Z = -0.30 SD`) and classifies the infant (**SGA <10th**, **AGA 10th–90th**, or **LGA >90th**).

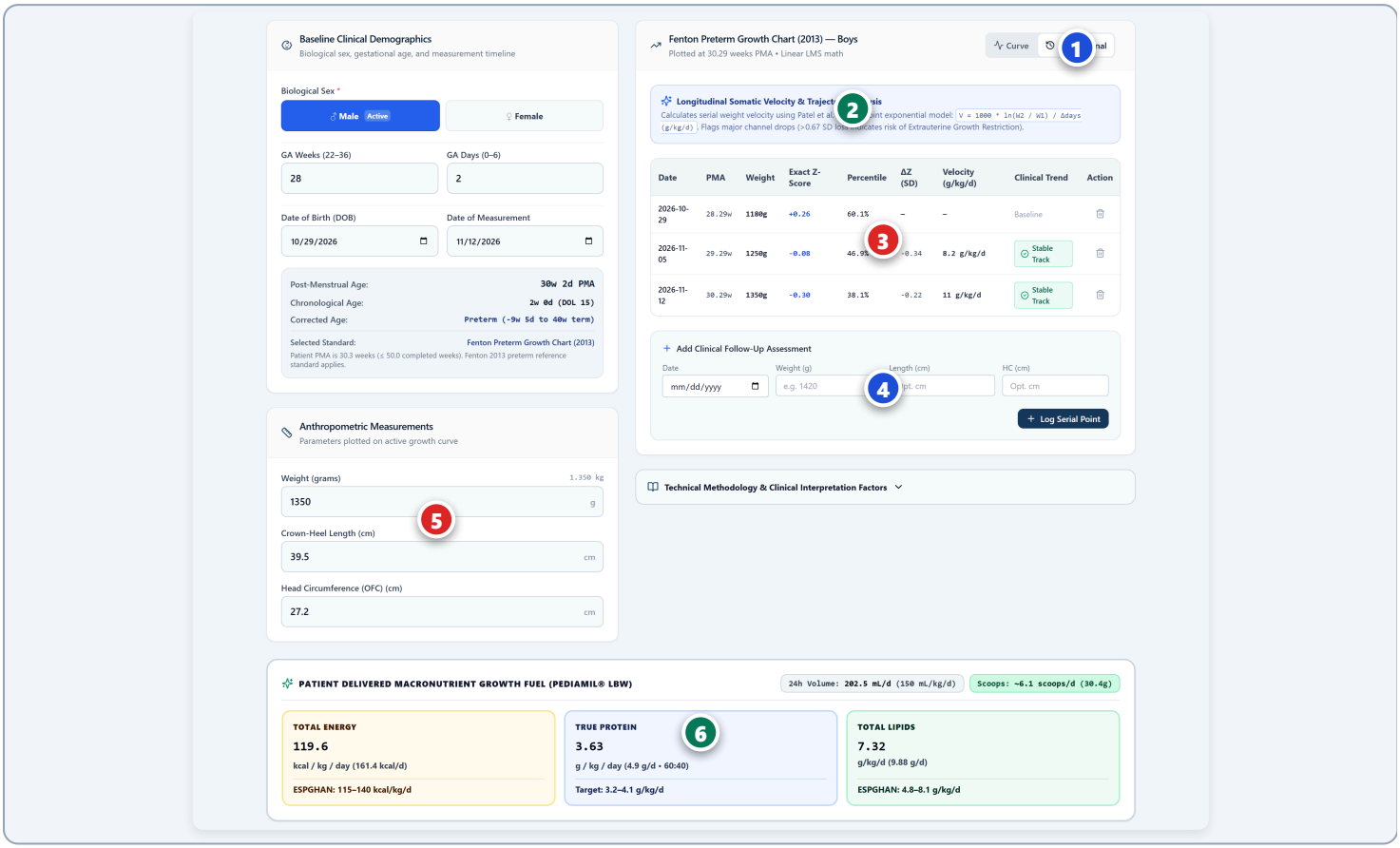
Cole & Green 1992 • Stat Med PMID: 1518992 **Chou et al. 2020 PediTools • J Pediatr PMID: 32012066**

TAB 1 (PART B): SERIAL LONGITUDINAL VELOCITY (PATEL ET AL. 2005), EUGR ALERTS & FUEL SNAPSHOT

SCREEN TOUR ① – ⑥

HOW LONGITUDINAL TRACKING HELPS THE DOCTOR PREVENT EXTRAUTERINE GROWTH RESTRICTION (EUGR)

A single weight percentile on admission does not tell the whole story: a preterm infant who drops from the 60th percentile at birth to the 20th percentile at week 3 is suffering from **Extrauterine Growth Restriction (EUGR)** even though both points sit inside the "normal" 10th–90th band. Clicking the **Longitudinal** toggle in Tab 1 automatically calculates **2-point exponential somatic velocity (g/kg/day)** and flags any downward trajectory exceeding **−0.67 SD** (one major centile channel).



1 Curve vs. Longitudinal Mode Switcher

Located at the top-right of the growth chart card. Clicking **Longitudinal** switches from cross-sectional percentile plotting to serial multi-point trajectory analysis across the infant's entire NICU hospital stay.

2 Patel et al. (2005) 2-Point Exponential Velocity Engine

Computes daily weight gain velocity in **g/kg/day** using Patel's validated neonatal exponential formula rather than crude linear averaging:

$$\text{Velocity (g/kg/d)} = [\ 1000 \times \ln(W_2 / W_1)\ ] / (\text{Date}_2 - \text{Date}_1)$$

Target preterm catch-up velocity is **15–20 g/kg/day**.

Patel et al. 2005 • Pediatrics PMID: 16253578

3 Serial Trajectory Table & ΔZ-Score EUGR Surveillance

Displays each logged assessment's **Date**, **PMA** (e.g., `28.29w` → `29.29w` → `30.29w`), **Weight** (`1180g` → `1250g` → `1350g`), **Exact Z-Score**, **Percentile**, **ΔZ (SD change)**, and **Velocity (g/kg/d)**.

- **Clinical Trend Badge**: Shows `Stable Track` when ΔZ is within physiological bounds, or triggers an immediate **EUGR Risk Warning** if cumulative Z-score drops by **> 0.67 SD**.

4 Add Clinical Follow-Up Assessment Log

Allows the physician or dietitian to enter a follow-up **Date**, **Weight (g)**, optional **Length (cm)**, and optional **HC (cm)** and click **+ Log Serial Point** to append the new measurement and recalculate exponential velocity immediately.

5 Full Anthropometric Measurements Panel (Weight, Length, HC)

Shows all three anthropometric inputs clearly on the left column:

- **Weight (grams)**: `1350 g` (with live `1.350 kg` conversion badge).
- **Crown-Heel Length (cm)**: `39.5 cm` (for linear bone growth assessment).
- **Head Circumference / OFC (cm)**: `27.2 cm` (critical proxy for brain volume accretion and neurodevelopmental protection).

6 Patient Delivered Macronutrient Growth Fuel Snapshot

Positioned directly beneath the growth plotter so the doctor can correlate somatic growth velocity with current nutritional intake on the same screen:

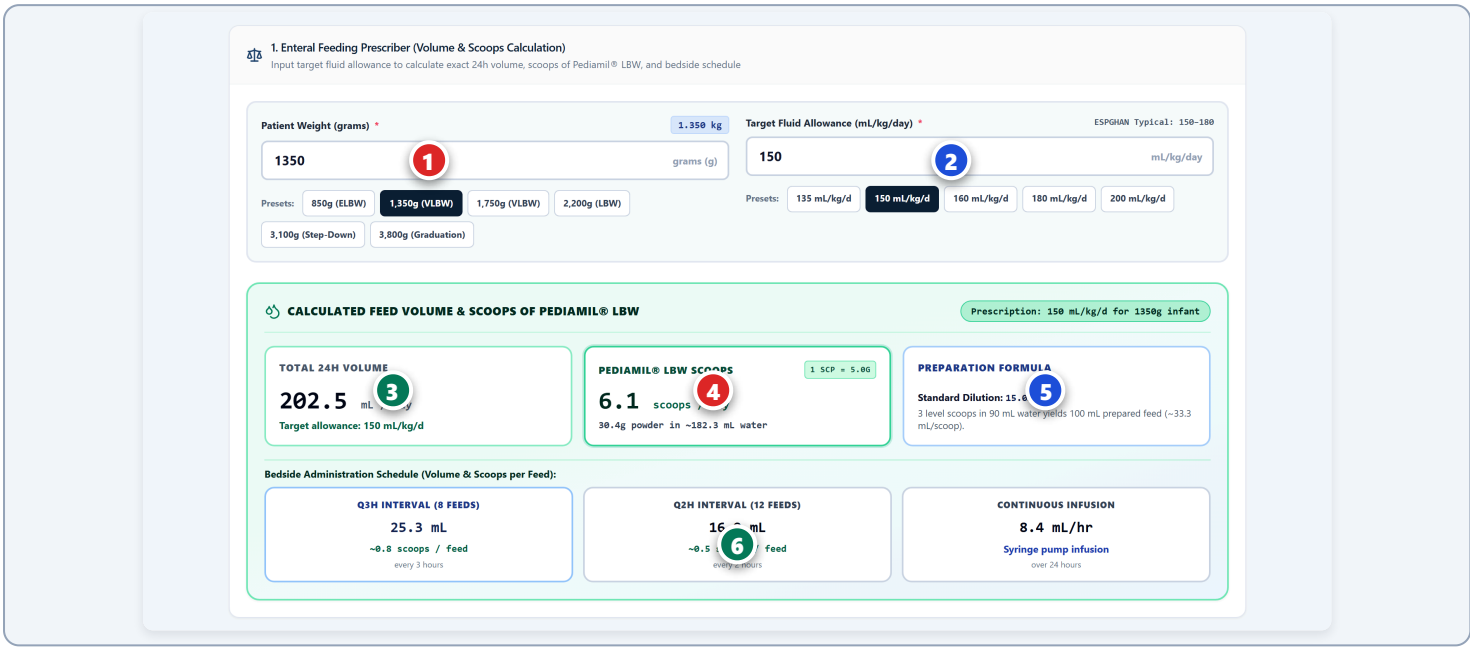
- Displays **24h Volume: 202.5 mL/d (150 mL/kg/d)** and **Scoops: ~6.1 scoops/d (30.4g)**.
- Shows **Total Energy (119.6 kcal/kg/d)**, **True Protein (3.63 g/kg/d • 60:40)**, and **Total Lipids (7.32 g/kg/d)** against ESPGHAN 2022 ranges.

TAB 2 (PART A): STEP 1 — ENTERAL FEEDING PRESCRIBER (24H VOLUME, PEDIAMIL® LBW SCOOPS & SCHEDULE)

SCREEN TOUR ① – ⑥

SIMPLIFYING BEDSIDE PRESCRIBING: FROM WEIGHT & FLUID ALLOWANCE TO EXACT VOLUME, SCOOPS & NURSING SCHEDULE

Doctors requested that the **Nutritional Feeds & Macro Engine** be intuitive, fast, and practical at the bedside—without requiring repeated data entry. In **Step 1**, the infant's weight is already carried over from Tab 1. From just **Weight (g)** and **Fluid Allowance (mL/kg/day)**, the engine immediately calculates **Total 24h Volume (mL/day)**, **exact Scoops of Pediamil® LBW per day**, **Sterile Water required (mL/day)**, and the **Bedside Administration Schedule (q3h, q2h, and Continuous Infusion)**.



1 Patient Weight Input & 1-Tap Clinical Weight Presets

Synchronized automatically with Tab 1 (`1350 g = 1.350 kg`). Includes 1-tap clinical presets for rapid ward rounds:

- `850g (ELBW)`, `1,350g (VLBW)`, `1,750g (VLBW)`, `2,200g (LBW)`, `3,100g (Step-Down)`, and `3,800g (Graduation to Pediamil® 1)`.
- Built-in safety gate blocks weights < `400 g` and automatically transitions infants > `3,500 g` to term **Pediamil® 1**.

2 Target Fluid Allowance Input & ESPGHAN 2022 Presets

Enter the daily enteral fluid allowance in **mL/kg/day** or tap one of the standard NICU presets: `135`, `150`, `160`, `180`, or `200 mL/kg/d`.

- **ESPGHAN 2022 Guideline**: Typical enteral fluid intake for stable growing preterm infants is **150–180 mL/kg/day** (up to `200 mL/kg/day` conditionally).

Embleton et al. ESPGHAN 2022 • PMID: 36705703

3 Total 24-Hour Prescribed Feed Volume (mL/day)

Instantly computes the exact 24-hour prepared formula volume:

`Total 24h Volume = (Weight_g ÷ 1000) × Fluid Allowance`

For a `1,350 g` infant at `150 mL/kg/day`: `1.350 × 150 = 202.5 mL/day`.

4 Pediamil® LBW Daily Scoops & Powder Mass

Converts the 24-hour volume directly into **Pediamil® LBW scoops** and **grams of powder** using the verified product specification (`15.0% w/v` dilution; `1 level scoop = 5.0 g powder`):

`Powder (g/d) = Volume × 0.15 = 30.4 g/day`

`Scoops/day = Powder ÷ 5.0 g = 6.1 scoops/day`

5 Standard Reconstitution & Sterile Water Rule

Displays the exact milk-kitchen reconstitution instructions:

- **Standard Dilution (15.0% w/v)**: Mix **3 level scoops (15.0 g powder)** in **90 mL sterile water** (>70°C, cooled) to yield **100 mL prepared feed** (~`33.3 mL feed per scoop`).
- For `202.5 mL/day`: requires **~182.3 mL sterile water** + **30.4 g powder (6.1 scoops)**.

WHO/FAO PIF Guidelines • ISBN 978-92-4-159541-4

6 Bedside Administration Schedule (Volume & Scoops per Feed)

Automatically divides the 24-hour order into three standard NICU administration modalities:

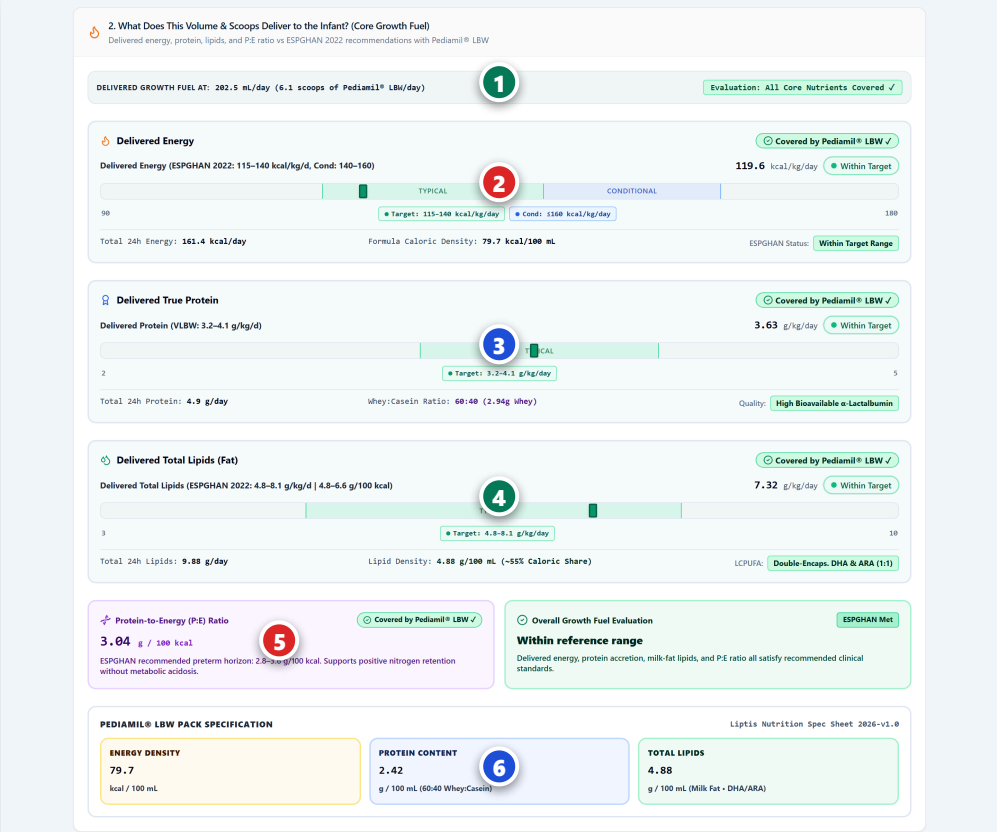
- **q3h Interval (8 feeds/day)**: `25.3 mL / feed` (~`0.8 scoops / feed` every 3 hours).
- **q2h Interval (12 feeds/day)**: `16.9 mL / feed` (~`0.5 scoops / feed` every 2 hours).
- **Continuous Infusion (24 hours)**: `8.4 mL / hour` via syringe pump.

INFANT WEIGHT	FLUID ALLOWANCE	TOTAL 24H VOLUME	PEDIAMIL® LBW SCOOPS/DAY	POWDER & WATER REQUIRED	Q3H BOLUS (8 FEEDS/D)	Q2H BOLUS (12 FEEDS/D)
1,000 g (1.00 kg)	150 mL/kg/day	**150.0 mL/day**	**4.5 scoops/day**	22.5 g powder + 135.0 mL water	18.8 mL (~0.6 scp/feed)	12.5 mL (~0.4 scp/feed)
1,350 g (1.35 kg)	150 mL/kg/day	**202.5 mL/day**	**6.1 scoops/day**	30.4 g powder + 182.3 mL water	25.3 mL (~0.8 scp/feed)	16.9 mL (~0.5 scp/feed)
1,750 g (1.75 kg)	160 mL/kg/day	**280.0 mL/day**	**8.4 scoops/day**	42.0 g powder + 252.0 mL water	35.0 mL (~1.1 scp/feed)	23.3 mL (~0.7 scp/feed)
2,200 g (2.20 kg)	160 mL/kg/day	**352.0 mL/day**	**10.6 scoops/day**	52.8 g powder + 316.8 mL water	44.0 mL (~1.3 scp/feed)	29.3 mL (~0.9 scp/feed)

TAB 2 (PART B): STEP 2 — WHAT DOES THIS VOLUME & SCOOPS DELIVER? (CORE GROWTH FUEL) SCREEN TOUR ① – ⑥

INSTANT VISUAL VERIFICATION OF ENERGY, TRUE PROTEIN (60:40), TOTAL LIPIDS & P:E RATIO AGAINST ESPGHAN 2022

Immediately below the volume and scoop calculator, **Step 2** answers the doctor's primary clinical question: "*At this prescribed daily volume and scoop count, how much Energy, Protein, and Fat is actually delivered to the infant per kg per day, and does Pediamil® LBW cover the ESPGHAN 2022 recommendation?*" Note that per clinical team alignment, carbohydrate range gauges have been removed so the interface focuses cleanly on **Energy, Protein, Lipids, and Micronutrients**.



1 Delivered Growth Fuel Summary Banner

Displays the active prescription context (`Delivered Growth Fuel at: 202.5 mL/day [6.1 scoops of Pediamil® LBW/day]`) and the dynamic overall badge (`Evaluation: All Core Nutrients Covered ✓` when within ESPGHAN 2022 targets).

2 Delivered Energy Gauge (kcal/kg/day)

Calculates `Energy = (Volume × 79.7 kcal/100mL) ÷ Weight_kg`.

- At `150 mL/kg/d`: delivers **119.6 kcal/kg/day** (`161.4 kcal/day` total).
- Evaluated against **ESPGHAN 2022 Target: 115–140 kcal/kg/day** (Conditional catch-up: `140–160 kcal/kg/day`) → **Covered by Pediamil® LBW ✓**.

3 Delivered True Protein Gauge (g/kg/day • 60:40 Whey:Casein)

Calculates `Protein = (Volume × 2.42 g/100mL) ÷ Weight_kg`.

- At `150 mL/kg/d`: delivers **3.63 g/kg/day** (`4.9 g/day` total; `60:40 Whey:Casein = 2.94g Whey` rich in bioavailable α-lactalbumin).
- Evaluated against **ESPGHAN 2022 Target: 3.5–4.0 g/kg/day** (acceptable `3.2–4.1`, conditional up to `4.5`) → **Covered by Pediamil® LBW ✓**.

4 Delivered Total Lipids / Fat Gauge (g/kg/day)

Calculates `Lipids = (Volume × 4.88 g/100mL) ÷ Weight_kg`.

- At `150 mL/kg/d`: delivers **7.32 g/kg/day** (`9.88 g/day` total; ~55% of total energy, natural milk fat + double-encapsulated 1:1 DHA & ARA).
- Evaluated against **ESPGHAN 2022 Target: 4.8–8.1 g/kg/day → Covered by Pediamil® LBW ✓**.

5 Protein-to-Energy (P:E) Ratio & Dynamic Status Evaluation

Computes `P:E Ratio = (Protein × 100) ÷ Energy = 3.04 g / 100 kcal`.

- Sits squarely inside the **ESPGHAN 2022 recommended 2.8–3.6 g/100 kcal** window, promoting lean mass accretion rather than excess adiposity while avoiding metabolic acidosis or elevated BUN.

6 Pediamil® LBW Official Pack Specification Strip

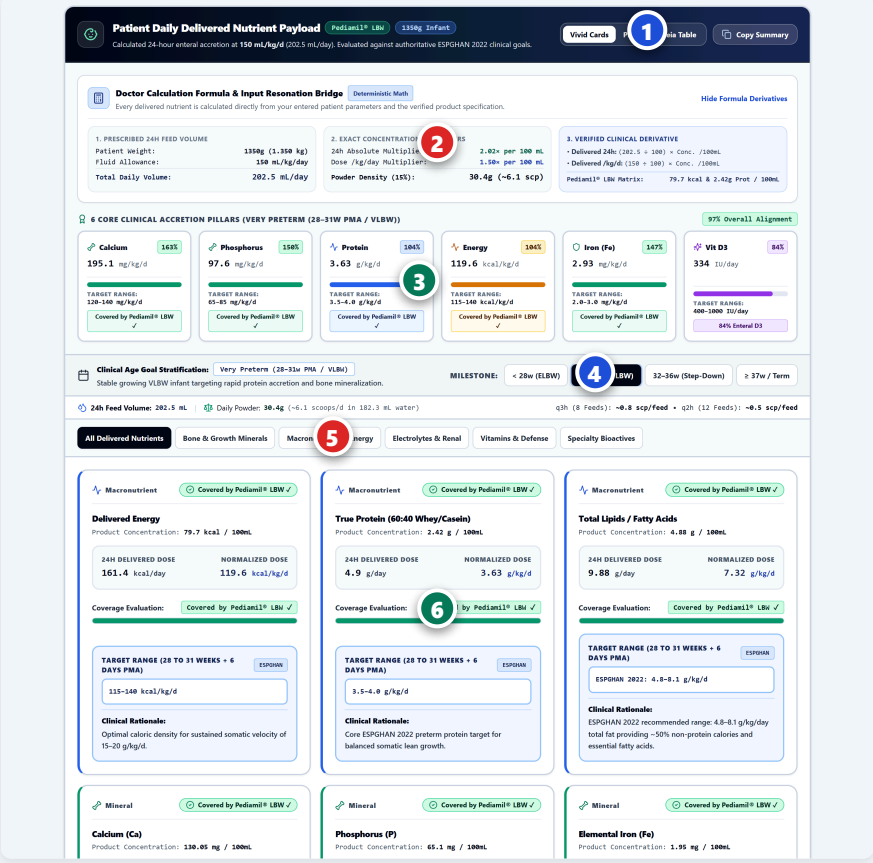
Displays the verified per-100 mL reference densities from the official Liptis Nutrition Specification Sheet (`Liptis-Spec-2026-v1.0`): **Energy Density: 79.7 kcal/100 mL • Protein Content: 2.42 g/100 mL (60:40 Whey:Casein) • Total Lipids: 4.88 g/100 mL (Milk Fat • DHA/ARA)**.

TAB 2 (PART C): STEP 3 — PATIENT DAILY DELIVERED NUTRIENT PAYLOAD & 6 ACCRETION PILLARS

SCREEN TOUR ① – ⑥

COMPLETE MACRO- & MICRONUTRIENT TRANSPARENCY: EVERY SINGLE INGREDIENT CHECKED AGAINST ESPGHAN 2022

Step 3 (Patient Daily Delivered Nutrient Payload) shows the doctor exactly how much of every macro- and micronutrient the infant receives per 24 hours and per kg/day—including **Calcium, Phosphorus, Iron, Vitamin D3, Sodium, Potassium, Chloride, Magnesium, Zinc, Copper, Selenium, Vitamins A/E/K1/C/B-complex, DHA, ARA, Inositol, Taurine, L-Carnitine, Choline, and Nucleotides**—and flags whether each ingredient is **Covered by Pediamil® LBW ✓**.



1 View Switcher (Vivid Cards vs. Pharmacopoeia Table) & Copy

Allows the doctor to switch between **Vivid Cards** (visual progress cards for ward rounds) and **Pharmacopoeia Table** (complete 20+ ingredient tabular audit with atomic `mmol/kg/d` conversions), plus 1-click **Copy Summary** for hospital EMR notes.

2 Doctor Calculation Formula & Input Resonation Bridge

Shows the exact deterministic math connecting the doctor's inputs to every nutrient card:

- **24h Absolute Multiplier**: `Volume ÷ 100 = 2.025× per 100 mL`
- **Dose /kg/day Multiplier**: `Fluid Allowance ÷ 100 = 1.50× per 100 mL`
- **Powder Density (15%)**: `30.4 g (~6.1 scoops)`.

3 6 Core Clinical Accretion Pillars Dashboard

Highlights the 6 primary drivers of preterm skeletal mineralization and lean tissue growth at `150 mL/kg/d`:

- **Calcium**: `195.1 mg/kg/d` (163% of 120 mg min • **Covered** ✓)
- **Phosphorus**: `97.6 mg/kg/d` (150% of 65 mg min • **Covered** ✓)
- **Protein**: `3.63 g/kg/d` (104% • **Covered** ✓) & **Energy**: `119.6 kcal/kg/d` (104% • **Covered** ✓)
- **Iron (Fe)**: `2.93 mg/kg/d` (147% • **Covered** ✓) & **Vit D3**: `334 IU/day` (84% of 400 IU/d absolute floor).

4 Clinical Age Goal Stratification Selector

Lets the clinician inspect nutrient alignment across 4 developmental milestones: `< 28w (ELBW)`, `28–31w (VLBW)`, `32–36w (Step-Down)`, and `≥ 37w / Term`, automatically adjusting target reference horizons.

5 Physiological Subsystem Filter Tabs

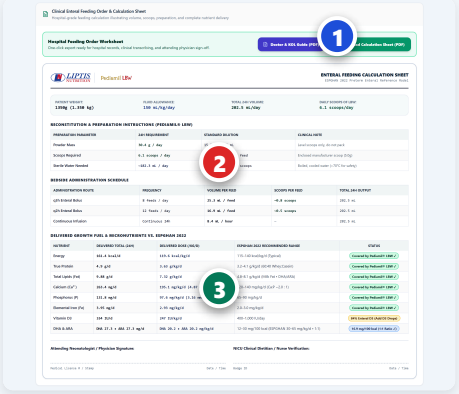
Filters the nutrient cards into 6 clinical categories: **All Delivered Nutrients**, **Bone & Growth Minerals** (Ca, P, Mg, Vit D3), **Macronutrients & Energy**, **Electrolytes & Renal** (Na, K, Cl), **Vitamins & Defense**, and **Specialty Bioactives** (DHA, ARA, Inositol, Nucleotides).

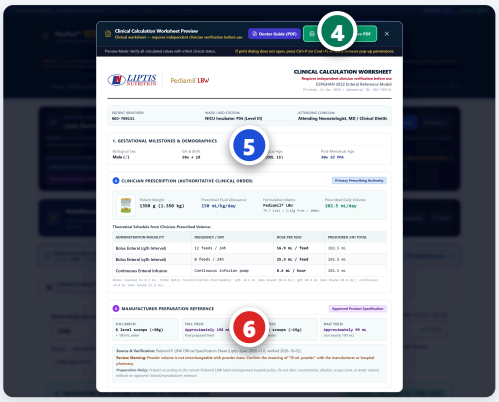
6 Individual Nutrient Cards with ESPGHAN Coverage Badges

Every card displays: (a) **Product Concentration per 100 mL**, (b) **24h Delivered Dose**, (c) **Normalized Dose (/kg/d)**, (d) **ESPGHAN 2022 Target Range**, (e) **Coverage Evaluation Badge (Covered by Pediamil® LBW ✓)**, and (f) **Clinical Rationale**.

STEP 4: EMBEDDED CALCULATION SHEET, HOSPITAL PDF MODAL & SCIENTIFIC SOURCES

SCREEN TOUR ① – ④ & MASTER REFERENCE REGISTRY





1 1-Click PDF Export Bar

Download the **Doctor & KOL Guide (PDF)** or open the **Printable Calculation Sheet (PDF)** directly from the bedside view.

2 Embedded Reconstitution & Schedule

Shows 24h Powder (`30.4g`), Scoops (`6.1 scp`), Water (`~182.3 mL`), and `q3h` / `q2h` / `Continuous` dosing without opening a modal.

3 ESPGHAN Table & Dual Signatures

Embedded table of delivered nutrients vs. ESPGHAN 2022 plus sign-off lines for the **Attending Neonatologist** and **NICU Dietitian/Nurse**.

4 Hospital Worksheet Print / Save PDF

Triggers A4 hospital worksheet printing (`window.print()`) with zero browser clutter for placement in the infant's physical medical chart.

5 Section 1 & A: Ages + Prescription

Prints Biological Sex, GA at Birth (`28w+2d`), CA (`2w 0d`), PMA (`30w 2d`), Prescribed Volume (`202.5 mL/d`), and feed schedule.

6 Section B: Batch Preparation Guide

Prints Full Batch (`6 scoops [~30g] + 180 mL water = ~198 mL`) and Half Batch (`3 scoops [~15g] + 90 mL water = ~99 mL`) milk-kitchen cards.

NUTRIENT / PARAMETER	PEDIAMIL® LBW (/100G POWDER)	PER 100 ML FEED (15% W/V)	DELIVERED @ 150 ML/KG/D	ESPGHAN 2022 PRETERM TARGET	CLINICAL COVERAGE STATUS
Energy	531 kcal	**79.7 kcal**	**119.6 kcal/kg/day**	115–140 kcal/kg/d (Cond: 140–160)	**Covered by Pediamil® LBW ✓**
True Protein (60:40 Whey:Casein)	16.1 g	**2.42 g**	**3.63 g/kg/day** (P:E 3.04)	3.5–4.0 g/kg/d (3.2–4.1; Cond: 4.5)	**Covered by Pediamil® LBW ✓**
Total Lipids (Milk Fat + MCT)	32.5 g	**4.88 g**	**7.32 g/kg/day**	4.8–8.1 g/kg/day	**Covered by Pediamil® LBW ✓**
Calcium (Ca^{2+}) & Phosphorus (P)	867 mg Ca / 434 mg P	**130.05 mg / 65.10 mg**	**195.1 / 97.6 mg/kg/d** (Ca:P 2:1)	Ca: 120–140 | P: 65–90 mg/kg/d	**Covered by Pediamil® LBW ✓**
Elemental Iron (Fe) & Zinc (Zn)	13.0 mg Fe / 7.0 mg Zn	**1.95 mg Fe / 1.05 mg Zn**	**2.93 mg Fe / 1.58 mg Zn /kg/d**	Fe: 2.0–3.0 | Zn: 1.4–2.5 mg/kg/d	**Covered by Pediamil® LBW ✓**
Electrolytes (Na / K / Cl / Mg)	265 / 555 / 400 / 53 mg	**39.8 / 83.3 / 60.0 / 7.95 mg**	**2.60 / 3.20 / 2.54 mmol/kg/d**	Na 1.5–3 | K 2.3–4.6 | Cl 1.5–3 mmol	**Covered by Pediamil® LBW ✓**
Vitamin D3 (Cholecalciferol)	1,100 IU (27.5 µg)	**165 IU (4.13 µg)**	**247.5 IU/kg/d** (334 IU/d @ 1.35kg)	400–1,000 IU/day (absolute daily)	**≥400 IU/d at ≥243 mL/d (Add D3 Drops if <243 mL/d)**
DHA & ARA (1:1 Double-Encaps.)	90 mg DHA / 90 mg ARA	**13.5 mg DHA / 13.5 mg ARA**	**20.25 mg/kg/d** (16.9 mg/100 kcal)	Codex/EFSA 12–30 mg/100kcal (ESPGHAN 30–65)	**Meets 16.9 mg/100kcal & 1:1 Ratio ✓**

CLINICAL DOMAIN	PRIMARY PEER-REVIEWED SCIENTIFIC REFERENCE	PUBMED PMID / IDENTIFIER
1. Preterm Enteral Nutrition	Embleton ND, Moltu SJ, Lapillonne A, et al. *Enteral Nutrition in Preterm Infants (2022): A Position Paper From the ESPGHAN Committee on Nutrition and Invited Experts.* J Pediatr Gastroenterol Nutr (JPGN). 2023;76(2):248–268.	**PMID: 36705703** DOI: 10.1097/MPG.0000000000003642
2. Preterm Growth Curves (22–50w)	Fenton TR, Kim JH. *A systematic review and meta-analysis to revise the Fenton growth chart for preterm infants.* BMC Pediatrics. 2013;13:59.	**PMID: 23601190** PMCID: PMC3637477
3. Term Growth Standards (0–24m)	WHO Multicentre Growth Reference Study Group. *WHO Child Growth Standards based on length/height, weight and age.* Acta Paediatr Suppl. 2006;450:76–85.	**PMID: 16817681** ISBN: 92-4-154693-X
4. Box-Cox LMS Z-Score Method	Cole TJ, Green PJ. *Smoothing reference centile curves: the LMS method and penalized likelihood.* Statistics in Medicine. 1992;11(10):1305–1319.	**PMID: 1518992** DOI: 10.1002/sim.4780111005
5. Exponential Weight Velocity	Patel AL, Engstrom JL, Meier PP, Kimura RE. *Accuracy of methods for calculating postnatal growth velocity for extremely low birth weight infants.* Pediatrics. 2005;116(6):1466–1473.	**PMID: 16253578** DOI: 10.1542/peds.2004-1699
6. PediTools LMS Validation	Chou JH, Roumiantsev S, Singh R. *PediTools Electronic Growth Chart Calculators: Applications in Clinical Care, Research, and Quality Improvement.* J Med Internet Res. 2020;22(1):e16204.	**PMID: 32012066** PMCID: PMC7016617
7. PIF Reconstitution & Atomic Weights	WHO/FAO. *Safe preparation, storage and handling of powdered infant formula: guidelines.* Geneva: WHO (2007) • IUPAC Commission on Isotopic Abundances and Atomic Weights (CIAAW).	**ISBN: 978-92-4-159541-4** IUPAC Standard Atomic Weights